

Product Bulletin

GEOSPATIAL

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PUBLIC

Enabling Dust Filter in TMI for Trimble MX60

This Product Bulletin describes the idea of the field dust filter that can be enabled in TMI by Trimble MX60 users. It describes scenarios when the dust filter can be used by the mobile mapping system operator and provides the requirement for this tool being utilized properly.

Dust Filter principles

When it might be useful

The dust filtering option has been implemented for MX60 users to avoid capturing of large volumes of noise and to help system operators to collect as many surface-related points as possible in extremely dusty environments such as unpaved roads and open pit mines when it is not possible to mount the mobile mapping system in front of the vehicle to avoid large volumes of dust being captured.

It is not foreseen with the current design to use dust filtering options in the environment of paved roads such as urban canyons and intercity infrastructure, etc.

How does it work

The dust filter can be activated by the Trimble MX60 system operator in the Capture setting of the Mission Presets tab like it is shown on the picture below:



Pic 1. Enabling Dust Filter in Capture Settings of MX60 Mission

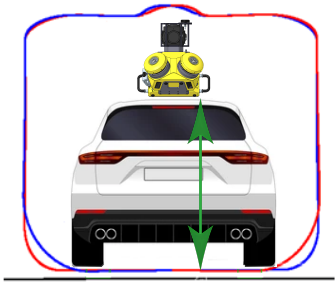
<https://surveypartners.trimble.com/>

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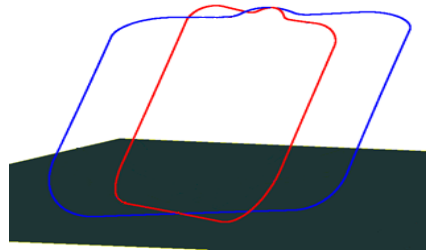
When activating the dust filter option in TMI “filter box area” is being defined, based on the measured MX60 system installation height information. Design of the “filter box area” implies a variety of vehicles mobile mapping systems can be mounted on.

All objects inside the “filter box area” will not be measured by the Trimble MX60 laser unit.

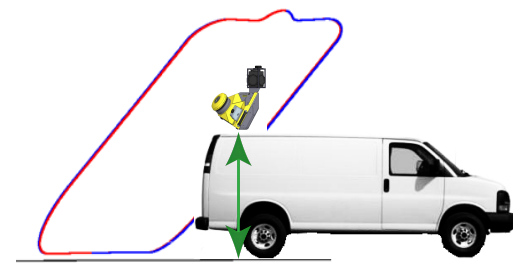
You can see “filter box area” illustrated on the pictures below:



Pic 2. Illustration of back view of “filter box area”



Pic 3. Illustration of 3D view of “filter box area”



Pic 4. Illustration of side view of “filter box area”

- Red - “filter box-area” Laser 1
Blue - “filter box-area” Laser 2
↕ - Trimble MX60 vertical installation height (to the external reference cross on the right side of the mounting rack)

What must be considered when enabling Dust Filter option in TMI

1. The Dust filter shall be enabled **in recommended scenarios only**, i.e. extremely dusty environments of unpaved roads or open pit mines.
2. **Trimble MX60 installation height** is a critical value that needs to be provided to apply the dust filter properly. It is important to measure installation height strictly vertically, hence being perpendicular to the XY plane of your Trimble MX60 system with the minimum accuracy of 2-3 cm. In case the vehicle is going to be heavily loaded during the data collection or it has very smooth suspensions, it needs to be considered when measuring the installation height parameter.¹

Important: *If the installation height provided by the MX60 system operator is higher than it will be in reality, it will lead to data loss (larger mask design assumed for the actual vehicle => mask touches the ground surface).*

¹ Detailed information about how to measure your system installation height can be found in [Trimble MX60 User Guide](https://geospatial.trimble.com/UserGuide)

3. It is recommended to operate your Trimble MX60 system with **1000 kHz laser effective measurement rate** instead of 500 kHz to achieve a better result (recommended laser mode can be set in Mission Presets/Capture settings).
4. It is recommended to **reduce driving speed** as much as possible in dusty environments.
5. It is recommended to avoid harsh vehicle maneuvers as much as possible.
6. If your environment changes during the mission, you can "reconfigure mission" using capture settings, enabling or disabling the dust filter as needed. Capture settings enabling and disabling the dust filter in this case would need to be prepared in advance.

Other Notes:

Trimble MX60 system operator can see the result of enabled dust filter option in TMI laser waterfall view. In case the wrong MX system installation height has been applied, data gaps can be visualized immediately.

For more information

For any questions concerning technical details, please contact the mobile mapping support team mx_support@trimble.com

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